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Improvement of third-order WENO-Z scheme at critical points

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ABSTRACT

In this paper, we present an improved third-order WENO-Z scheme to improve the order of convergence at critical points. This scheme is constructed through slightly modifying the local smoothness indicators of the conventional WENO-Z scheme with the way of Taylor expansion for third-order convergence. The present scheme is proved to be close to third-order by several standard accuracy tests. The performance enhancement of the WENO scheme through this modification is verified on a variety of one-dimensional and two-dimensional standard numerical experiments. Numerical results indicate that the present scheme provides better results in comparison with the earlier third-order WENO schemes like WENO-JS3, WENO-Z3.

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1. Introduction

Weighted essentially nonoscillatory (WENO) schemes are a popular class of numerical schemes for hyperbolic conservation laws. They were first proposed in [1] and then improved by Jiang and Shu [2]. However, in [3], the authors find that this scheme fails to achieve the optimal accuracy at critical points. Then, fifth-order WENO-M scheme [3] and fifth-order WENO-Z scheme [4] are proposed to recover the designed order at critical points. Castro et al. [5] developed a general formula for the higher order smoothness indicators and extended the WENO-Z scheme to all odd orders of accuracy.

Several approaches have been considered for the improvement of the WENO-Z scheme convergence order in recent years. For the high-order WENO schemes (at least fifth-order), Don and Borges [6] discussed the theoretical analysis on the accuracy of the high-order WENO-Z schemes based on the parameters p and ε introduced in the formulation of nonlinear weights. Youngsoo Ha et al. [7] present a new smoothness indicator that evaluates the local smoothness of a function inside of a stencil. The corresponding weighted essentially non-oscillatory (WENO) finite difference scheme can provide the fifth convergence order in smooth regions, especially at critical points where the first derivative vanishes. Chang Ho Kim et al. [8] further improve the smoothness indicator of WENO-NS and suggest new nonlinear weights by simplifying WENO-NS weights. The proposed WENO scheme provides the fifth-order accuracy, even at critical points. A local smoothness indicator, which is based on Lagrangian interpolation polynomials, is introduced in [9,10] and this method is extended to high-order. Shen et al. [11] proposed a multi-step method by introducing fourth-order fluxes combined with high order smoothness indicator to improve the accurate near discontinuities for the fifth-order WENO scheme. Later on, Yankai Ma et al. [12] present an efficient procedure for overcoming the deficiency of weighted essentially nonoscillatory schemes near discontinuities up to ninth-order accurate multistep methods are devised while maintaining stability and the essentially non-oscillatory behavior. For the third-order WENO schemes, Yamaleev et al. [13] developed a new third-order Energy Stable

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Weighted Essentially Non-Oscillatory (ESWENO) finite difference scheme based on new weight functions for scalar and vector hyperbolic equations with piecewise continuous initial conditions. Wu and Zhao [14] proposed an improved third-order WENO-Z scheme (WENO-N3) through constructing a new reference smoothness indicator by a linear combination of the candidate and global smoothness indicator. In order to achieve the desired order of convergence at the critical point, Wu et al. [15] later introduced the WENO-NP3 scheme through constructing a new reference smoothness indicator by the nonlinear combination of the candidate and global smoothness indicators. Very recently, Gande et al. [16] have proposed a new reference smoothness indicator by the linear combination of the first derivative information of the local and global stencils to achieve desired order of convergence at critical points based on the works in [14,15].

From above status of research, it can be seen that improving the accuracy at critical points can improve the performance of the conventional WENO-Z scheme. In the present work, we have devised an improved WENO-Z scheme achieving optimal order near critical points.

The rest of this paper is organized as follows. In Section 2, the conventional WENO-JS3 scheme, WENO-Z3 scheme, and the proposed WENO-PZ3 scheme are introduced. Series numerical examples are provided in Section 3. Concluding remarks are given in Section 4.

2. Numerical methods

For the hyperbolic conservation law in the form

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0 \tag{1}$$

the flux function $f(u)$ can be split into two parts as $f(u) = f^+(u) + f^-(u)$ with $df^+(u)/du \geq 0$ and $df^-(u)/du \leq 0$. The semi-discretization form of Eq. (1) can be written as

$$\frac{du_i(t)}{\partial t} = -\frac{1}{h}\left(\overline{f}_{i+1/2} - \overline{f}_{i-1/2}\right) \tag{2}$$

here h denotes the uniform grid size. The numerical flux is $\overline{f}_{i+1/2} = \overline{f}^+_{i+1/2} + \overline{f}^-_{i+1/2}$. In this paper, only the positive part $\overline{f}^+_{i+1/2}$ is described, and the superscript '+' is dropped for simplicity.

2.1. WENO-JS3 scheme

The third-order WENO scheme [2] (WENO-JS3) use a convex combination of the candidate numerical fluxes to reconstruct the final flux

$$\overline{f}_{i+1/2} = \omega_0 f_{0,i+1/2} + \omega_1 f_{1,i+1/2} \tag{3}$$

where $f_{0,i+1/2}, f_{1,i+1/2}$ are the second-order fluxes defined on the two stencils $\{x_{i-1}, x_i\}$, $\{x_i, x_{i+1}\}$ as follows:

$$f_{0,i+1/2} = -\frac{1}{2}f_{i-1} + \frac{3}{2}f_i, f_{1,i+1/2} = \frac{1}{2}f_i + \frac{1}{2}f_{i+1}. \tag{4}$$

The classical weight functions ω_0 and ω_1 proposed by Jiang and Shu in [2] are given by

$$\omega_k = \frac{\alpha_k}{\sum_{s=0}^{1}\alpha_s}, \alpha_k = \frac{d_k}{(\varepsilon + \beta_k)^2}, k = 0, 1 \tag{5}$$

where the parameter ε in Eq. (6) is set to be 10^{-6} in the present work, as recommended in [2]. The optimal weights d_k, with the values

$$d_0 = \frac{1}{3}, d_1 = \frac{2}{3}. \tag{6}$$

The smoothness indicators are as follow

$$\beta_0 = (f_{i-1} - f_i)^2, \beta_1 = (f_i - f_{i+1})^2. \tag{7}$$

2.2. WENO-Z3 scheme

For the third-order WENO-Z scheme (WENO-Z3 scheme), similar to the classical weights, the weight functions are given by Eq. (5), whereas, the function α_k proposed in [6] are different from their classical counterparts and defined as

$$\alpha_k = d_k\left(1 + \frac{\tau_Z}{\beta_k + \varepsilon}\right), k = 0, 1, \tag{8}$$

$$\tau_Z = |\beta_0 - \beta_1|, \varepsilon = 10^{-40}. \tag{9}$$

2.2.1. Accuracy of WENO-Z3 scheme at non-critical points

Now, we present a short derivation to investigation on the accuracy of the WENO-Z3 scheme at the non-critical points in the smooth flow fields with the way of Taylor expansion.

The smoothness indicators of the candidate stencils in Eq. (7) can be expanded as a Taylor series at x_i

$$\begin{cases} \beta_0 = f_i'^2 h^2 - f_i' f_i'' h^3 + \left(\frac{1}{4} f_i''^2 + \frac{1}{3} f_i' f_i'''\right) h^4 - \frac{f_i' f_i''''}{12} h^5 - \frac{f_i'' f_i'''}{6} h^5 + O\left(h^6\right), \\ \beta_1 = f_i'^2 h^2 + f_i' f_i'' h^3 + \left(\frac{1}{4} f_i''^2 + \frac{1}{3} f_i' f_i'''\right) h^4 + \frac{f_i' f_i''''}{12} h^5 + \frac{f_i'' f_i'''}{6} h^5 + O\left(h^6\right). \end{cases} \tag{10}$$

At the non-critical points ($f_i' \neq 0, f_i'' \neq 0$), substituting Eqs. (9) and (10) into Eq. (8) disregarding the small parameter ε gives

$$\begin{aligned} \alpha_0 &= d_0\left(1 + \frac{\tau_Z}{\beta_0}\right) = d_0\left(1 + \frac{2f_i''}{f_i'} h \frac{1 - \frac{f_i''}{f_i'} h + \frac{f_i''}{f_i'} h + O\left(h^2\right)}{1 - \frac{f_i''}{f_i'} h + \frac{\left(\frac{1}{4} f_i''^2 + \frac{1}{3} f_i' f_i'''\right)}{f_i'^2} h^2 + O\left(h^3\right)}\right) \\ &= d_0\left(1 + \frac{2f_i''}{f_i'} h\left(1 + \frac{f_i''}{f_i'} h + O\left(h^2\right)\right)\right) = d_0\left[1 + \frac{2f_i''}{f_i'} h + \frac{2f_i''^2}{f_i'^2} h^2 + O\left(h^3\right)\right] \end{aligned} \tag{11}$$

$$\begin{aligned} \alpha_1 &= d_1\left(1 + \frac{\tau_Z}{\beta_1}\right) = d_1\left(1 + \frac{2f_i''}{f_i'} h \frac{1 + \frac{f_i''}{f_i'} h - \frac{f_i''}{f_i'} h + O\left(h^2\right)}{1 + \frac{f_i''}{f_i'} h + \frac{\left(\frac{1}{4} f_i''^2 + \frac{1}{3} f_i' f_i'''\right)}{f_i'^2} h^2 + O\left(h^3\right)}\right) \\ &= d_1\left(1 + \frac{2f_i''}{f_i'} h\left(1 - \frac{f_i''}{f_i'} h + O\left(h^2\right)\right)\right) = d_1\left[1 + \frac{2f_i''}{f_i'} h - \frac{2f_i''^2}{f_i'^2} h^2 + O\left(h^3\right)\right]. \end{aligned} \tag{12}$$

Then, with the weighting procedure in Eq. (5), we can obtain the following relation

$$\omega_0 = \frac{\alpha_0}{\alpha_0 + \alpha_1} = \frac{1}{3} + \frac{8}{9} \frac{f_i''^2}{f_i'^2} h^2 + O\left(h^3\right) \tag{13}$$

$$\omega_1 = \frac{\alpha_1}{\alpha_0 + \alpha_1} = \frac{2}{3} - \frac{8}{9} \frac{f_i''^2}{f_i'^2} h^2 + O\left(h^3\right). \tag{14}$$

Thus the numerical flux $\bar{f}_{i+1/2}$ in Eq. (2) is given by

$$\bar{f}_{i+1/2} = f_i + \frac{1}{2} f_i' h + \frac{1}{12} f_i'' h^2 + \frac{1}{12} f_i''' h^3 + \left(\frac{1}{144} f_i^{(4)} - \frac{4}{9} \frac{f_i''^3}{f_i'^2}\right) h^4 + O\left(h^5\right). \tag{15}$$

Similarly, the numerical flux $\bar{f}_{i-1/2}$ in Eq. (2) is given by

$$\bar{f}_{i-1/2} = f_i - \frac{1}{2} f_i' h + \frac{1}{12} f_i'' h^2 + \frac{1}{12} f_i''' h^3 - \left(\frac{11}{144} f_i^{(4)} + \frac{4}{9} \frac{f_i''^3}{f_i'^2}\right) h^4 + O\left(h^5\right). \tag{16}$$

Finally we get the following results

$$\frac{\bar{f}_{i+1/2} - \bar{f}_{i-1/2}}{h} = f_i' + \frac{1}{12} f_i^{(4)} h^3 + O\left(h^4\right) = f_i' + O\left(h^3\right). \tag{17}$$

From above results, it is clear that the WENO-Z3 scheme can achieve 3rd-order accuracy at the smooth flow field excluding the critical points.

2.2.2. Accuracy of WENO-Z3 scheme at critical points

At the first-order critical point ($f_i' = 0, f_i'' \neq 0, f_i''' \neq 0$), Eq. (10) can be written as the following form disregarding the small parameter ε

$$\begin{cases} \beta_0 = \frac{1}{4} f_i''^2 h^4 - \frac{f_i'' f_i'''}{6} h^5 + O\left(h^6\right), \\ \beta_1 = \frac{1}{4} f_i''^2 h^4 + \frac{f_i'' f_i'''}{6} h^5 + O\left(h^6\right). \end{cases} \tag{18}$$

Substituting Eq. (18), and Eq. (9) into Eq. (8) gives

$$\alpha_0 = d_0\left(1 + \frac{\tau_Z}{\beta_0}\right) = \frac{1}{3} + \frac{4}{9} \frac{f_i'''}{f_i''} h + \frac{8}{27} \frac{f_i'''^2}{f_i''^2} h^2 + O\left(h^3\right) \tag{19}$$

$$\alpha_1 = d_0\left(1+\frac{\tau_Z}{\beta_1}\right) = \frac{2}{3} + \frac{8}{9}\frac{f_i'''}{f_i''}h - \frac{16}{27}\frac{f_i'''^2}{f_i''^2}h^2 + O\left(h^3\right). \tag{20}$$

Then, with the weighting procedure in Eq. (5), we can obtain the following relation

$$\omega_0 = \frac{\alpha_0}{\alpha_0+\alpha_1} = \frac{1}{3} + \frac{32}{81}\frac{f_i'''^2}{f_i''^2}h^2 + O\left(h^3\right) \tag{21}$$

$$\omega_1 = \frac{\alpha_1}{\alpha_0+\alpha_1} = \frac{2}{3} - \frac{32}{81}\frac{f_i'''^2}{f_i''^2}h^2 + O\left(h^3\right). \tag{22}$$

Thus the numerical flux $\bar{f}_{i+1/2}$ in Eq. (2) is given by

$$\bar{f}_{i+1/2} = f_i + \frac{1}{2}f_i'h + \frac{1}{12}f_i''h^2 + \frac{1}{12}f_i'''h^3 + \left(\frac{1}{144}f_i^{(4)} - \frac{16}{81}\frac{f_i'''^2}{f_i''}\right)h^4 + O\left(h^5\right). \tag{23}$$

Similarly, the numerical flux $\bar{f}_{i-1/2}$ in Eq. (2) is given by

$$\bar{f}_{i-1/2} = f_i - \frac{1}{2}f_i'h + \frac{435}{2148}f_i''h^2 + O\left(h^3\right). \tag{24}$$

Finally we get the following results

$$\frac{\bar{f}_{i+1/2} - \bar{f}_{i-1/2}}{h} = f_i' - \frac{256}{2148}f_i''h + O\left(h^2\right) = f_i' + O\left(h\right). \tag{25}$$

Thus, the WENO-Z3 methods only achieve 1rd-order accuracy at critical points and this conclusion has been verified in [14,15] with a critical point problem test.

To achieve the desired convergence order for the scheme excluding the critical points, Wu et al. [15] introduced the WENO-NP3 scheme through constructing a new reference smoothness indicator by the nonlinear combination of the candidate and global smoothness indicators.

$$\begin{aligned} \alpha_k &= d_k\left(1+\frac{\tau_{NP}}{\beta_k+\varepsilon}\right), k=0,1, \\ \tau_{NP} &= \left|\frac{\beta_0+\beta_1}{2} - \beta_3\right|^p \end{aligned} \tag{26}$$

where β_3 is the smoothness indicator of the whole stencil $\{x_{i-1}, x_i, x_{i+1}\}$ and which is expressed as

$$\begin{aligned} \beta_3 &= \dot{\beta}_3 + \ddot{\beta}_3 \\ &= \left(\frac{1}{12}(f_{i-1}-2f_i+f_{i+1})^2 + \frac{1}{4}(f_{i-1}-f_{i+1})^2\right) + (f_{i-1}-2f_i+f_{i+1})^2. \end{aligned} \tag{27}$$

Shortly after that, based on the works in [15], Gande et al. [10] proposed another type of improved scheme through constructing a new reference smoothness indicator by the nonlinear combination of linear combination of the first derivative information of the local and global stencils. The improved reference smoothness indicator is denoted by τ_{NF} with the following form

$$\begin{aligned} \alpha_k &= d_k\left(1+\frac{\tau_{NF}}{\beta_k+\varepsilon}\right), k=0,1, \\ \tau_{NF} &= \left|\frac{\beta_0+\beta_1}{2} - \dot{\beta}_3\right|^p. \end{aligned} \tag{28}$$

In the present work, we denote this scheme as WENO-NF3 scheme. The parameter p is determined to be 3/2 by the sufficient condition for the third-order WENO scheme to achieve third-order accuracy as demonstrated in [5,6,13,14,16], which is given as follow

$$\omega_k - d_k = o\left(h^2\right). \tag{29}$$

2.3. The improved scheme

Following the works in the Refs. [15,16], we consider applying the above method directly to the local smoothness indicators in the function α_k of the conventional WENO-Z3 scheme discussed in Section 2.2. Therefore, the expression for the new reference smoothness indicator can be given by

$$\alpha_k = d_k\left(1+\frac{\tau_Z}{(\beta_k+\varepsilon)^p}\right), k=0,1 \tag{30}$$

$$\tau_Z = |\beta_0 - \beta_1|, \varepsilon = 10^{-40}. \tag{31}$$

2.3.1. Accuracy of the improved scheme at critical points

Here, we will firstly determine the convergence order at the first-order critical point ($f_i' = 0, f_i'' \neq 0, f_i''' \neq 0$) with the way of Taylor expansion. Substituting Eqs. (18) and (31) into Eq. (30) disregarding the small parameter ε gives

$$\begin{aligned}\alpha_0 &= d_0\left(1+\frac{\tau_Z}{\beta_0^p}\right)\\ &= \frac{1}{3}+\frac{1}{9}4^p\left(f_i''\right)^{1-2p}f_i'''h^{5-4p}+\frac{2}{27}p4^p\left(f_i''\right)^{-2p}\left(f_i'''\right)^2h^{6-4p}+O\left(h^{7-4p}\right)\end{aligned} \tag{32}$$

$$\begin{aligned}\alpha_1 &= d_1\left(1+\frac{\tau_Z}{\beta_1^p}\right)\\ &= \frac{2}{3}+\frac{2}{9}4^p\left(f_i''\right)^{1-2p}f_i'''h^{5-4p}-\frac{4}{27}p4^p\left(f_i''\right)^{-2p}\left(f_i'''\right)^2h^{6-4p}+O\left(h^{7-4p}\right).\end{aligned} \tag{33}$$

Then, with the weighting procedure in Eq. (5), the following relation can be obtained

$$\omega_0 = \frac{\alpha_0}{\alpha_0+\alpha_1} = \frac{1}{3}+\frac{8}{81}p4^p\left(f_i''\right)^{-2p}\left(f_i'''\right)^2h^{6-4p}+O\left(h^{7-4p}\right) \tag{34}$$

$$\omega_1 = \frac{\alpha_1}{\alpha_0+\alpha_1} = \frac{2}{3}-\frac{8}{81}p4^p\left(f_i''\right)^{-2p}\left(f_i'''\right)^2h^{6-4p}+O\left(h^{7-4p}\right). \tag{35}$$

Thus the numerical flux $\bar{f}_{i+1/2}$ in Eq. (2) is given by

$$\begin{aligned}\bar{f}_{i+1/2} =& f_i+\frac{1}{2}f_i'h+\frac{1}{12}f_i''h^2+\frac{1}{12}f_i'''h^3+\\ &\frac{1}{144}f_i^{(4)}h^4-\frac{4}{81}p4^p\left(f_i''\right)^{1-2p}\left(f_i'''\right)^2h^{8-4p}+O\left(h^5\right).\end{aligned} \tag{36}$$

Similarly, the numerical flux $\bar{f}_{i-1/2}$ in Eq. (2) is given by

$$\begin{aligned}\bar{f}_{i-1/2} =& f_i-\frac{1}{2}f_i'h+\frac{1}{12}f_i''h^2+\frac{1}{12}f_i'''h^3-\\ &\frac{11}{144}f_i^{(4)}h^4-\frac{2}{9}\frac{(4^p-36^p)}{9^p}f_i''^{3-2p}h^{6-4p}+O\left(h^5\right).\end{aligned} \tag{37}$$

Finally we get the following results

$$\frac{\bar{f}_{i+1/2}-\bar{f}_{i-1/2}}{h} = f_i'+\frac{1}{12}f_i^{(4)}h^3-\frac{4}{81}p4^p\left(f_i''\right)^{1-2p}\left(f_i'''\right)^2h^{7-4p}+\frac{2}{9}\frac{(4^p-36^p)}{9^p}f_i''^{3-2p}h^{5-4p}+O\left(h^4\right). \tag{38}$$

From Eq. (38), we can find that the parameter p should satisfy the following condition to achieve the optimal order of convergence at the first-order critical points

$$\begin{cases}5-4p\geq 3\\ 7-4p\geq 3\end{cases} \Rightarrow p\leq\frac{1}{2}. \tag{39}$$

For convenience, this proposed third-order WENO-Z scheme is denoted by WENO-PZ3 scheme. Then, in Section 2.3.2, we will give a short discussion on how to determine the final parameter p.

2.3.2. Determination of the final parameter p

Firstly, in this subsection, we investigate the behavior of the WENO-PZ3 scheme with three different choices of the parameter p: $p = 1/2, 3/4, 4/4$ through the following two typical numerical experiments. It should be noted that the WENO-PZ3 scheme with the parameter 4/4 is equivalent to the conventional WENO-Z3 scheme.

(1) *Gaussian–Square–Triangle–Ellipse linear test*

A challenging test case that contains a smooth combination of Gaussians, a square wave, a sharp triangle wave, and a half ellipse is calculated. The initial condition is as follow

$$u_0(x) = \begin{cases}\frac{1}{6}\left(G(x,\beta,z-\delta)+G(x,\beta,z+\delta)+4G(x,\beta,z)\right), & -0.8\leq x\leq -0.6\\ 1, & -0.4\leq x\leq -0.2\\ 1-|10(x-0.1)|, & 0\leq x\leq 0.2\\ \frac{1}{6}\left(F(x,\alpha,a-\delta)+F(x,\alpha,a+\delta)+4F(x,\alpha,a)\right), & -0.8\leq x\leq -0.6\\ 0, & otherwise\end{cases} \tag{40}$$

where

$$G(x,\beta,z)=e^{-\beta(x-z)^2},\ F(x,\alpha,a)=\sqrt{\max\left(1-\alpha^2\left(x-a^2\right),0\right)}. \tag{41}$$

The constants are taken as $a=0.5, z=-0.7, d=0.005, \alpha=10, \beta=1\mathrm{og}2/(36\delta^2)$.

(2) *Sod problem*

The initial conditions for the Sod problem are as follows

$$(\rho,u,p)=\begin{cases}(1,0,1), & 0\le x<0.5\\ (0.125,0,0.1), & 0.5\le x\le 1.\end{cases} \tag{42}$$

Fig. 1 plots the numerical results at $t=8$ with 200 grid points for the combination wave. Fig. 2 shows the density and enlarged portion near the discontinuity at $t=0.18$ with $N=200$ points for the shock tube problem. And, the 'exact' solution is the result computed on 10 000 points using the WENO-JS3 scheme. It can be observed from Figs. 1(a), 2(a), and 2(b) that the WENO-PZ3 schemes with the three parameters all achieve better resolution of waves than the WENO-JS3, WENO-Z3 schemes both at the critical points and the discontinuous. With the decrease of the parameter p, the less dissipative results can be obtained. However, the WENO-PZ3 scheme with parameter 1/2 gives obvious spurious oscillations. At the same time, we also give the simulation results of WENO-NP3, and WENO-NF3 schemes. It is clear from Figs. 1(b), 1(c), 2(c), and 2(d) that the WENO-NP3 scheme and the WENO-NF3 scheme both give obvious spurious oscillations for simulating above two cases. Thus, although the WENO-NP3 and the WENO-NF3 schemes really achieve third-order at critical points, however, the schemes do not consider eliminating the spurious oscillations when simulating some cases. In the following discussion, we will further give an investigation on how to choose the final parameter p for the proposed WENO-PZ3 scheme.

Secondly, we give an explanation for the better resolution results of WENO-PZ3 scheme than the conventional WENO-Z3 schemes and the spurious oscillations computed with smaller parameter $p=1/2$ by comparing the weights of less-smooth substencils between the WENO-PZ3 scheme and WENO-Z3 scheme.

Proposition. *Suppose S_C and S_D are two substencils of the same stencil, such that the solution is smoother at S_C than at S_D (meaning that $\beta_C<\beta_D$). Then* $\left[\frac{\omega_D}{\omega_C}\right]_p>\left[\frac{\omega_D}{\omega_C}\right]_1, (0<p<1)$.

Proof. Firstly we have the following relation disregarding the small parameter ε

$$\left[\frac{\omega_D}{\omega_C}\right]_p-\left[\frac{\omega_D}{\omega_C}\right]_1=\left[\frac{\alpha_D}{\alpha_C}\right]_p-\left[\frac{\alpha_D}{\alpha_C}\right]_1=\frac{d_D}{d_C}\left\{\frac{1+\frac{\tau}{\beta_D^p}}{1+\frac{\tau}{\beta_C^p}}-\frac{1+\frac{\tau}{\beta_D}}{1+\frac{\tau}{\beta_C}}\right\}. \tag{43}$$

Since $\beta_D>\beta_C$, we have

$$a=\beta_D, b=\beta_C\Rightarrow a>b>0.$$

Therefore

$$\frac{1+\frac{\tau}{a^p}}{1+\frac{\tau}{b^p}}-\frac{1+\frac{\tau}{a}}{1+\frac{\tau}{b}}=\frac{\frac{\tau\left[\left(b^p-a^p\right)\left(1-a^{p-1}\right)+a^p\left(b^{p-1}-a^{p-1}\right)\right]}{a^pb^p}+\frac{\tau^2ab^p\left(1-\left(\frac{a}{b}\right)^{p-1}\right)}{a^{p+1}b^{p+1}}}{\left(1+\frac{\tau}{b^p}\right)\left(1+\frac{\tau}{b}\right)}>0. \tag{44}$$

From Eqs. (43) and (44), we can obtain the conclusion

$$\left[\frac{\omega_D}{\omega_C}\right]_p>\left[\frac{\omega_D}{\omega_C}\right]_1, (0<p<1). \tag{45}$$

From Eq. (45), it can be indicated that the WENO-PZ3 scheme will be less dissipative than the conventional WENO-Z3 scheme, according to the conclusion that the scheme will become less dissipative with increased weights of less-smooth substencils in [17].

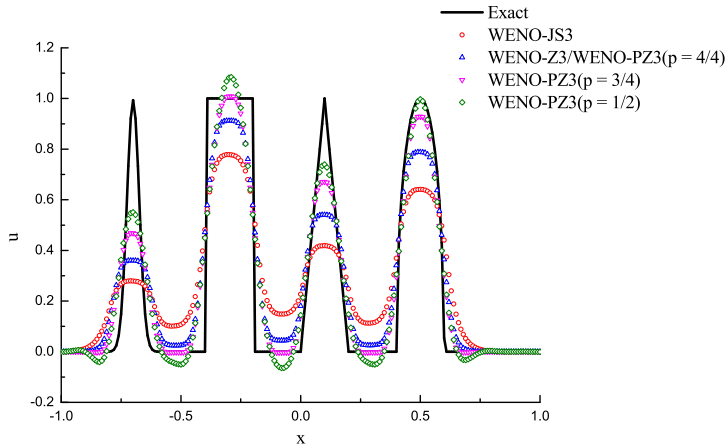
Similarly, we can obtain the following relation after some derivations

$$\frac{1+\frac{\tau}{a^{p_1}}}{1+\frac{\tau}{b^{p_1}}}-\frac{1+\frac{\tau}{a^{p_2}}}{1+\frac{\tau}{b^{p_2}}}=\frac{\frac{\tau\left[\left(b^{p_1}-a^{p_1}\right)\left(1-a^{p_1-p_2}\right)+a^{p_1}\left(b^{p_1-p_2}-a^{p_1-p_2}\right)\right]}{a^{p_1}b^{p_1}}+\frac{\tau^2a^{p_2}b^{p_1}\left(1-\left(\frac{a}{b}\right)^{p_1-p_2}\right)}{a^{p_1}b^{p_2}a^{p_2}b^{p_1}}}{\left(1+\frac{\tau}{b^{p_1}}\right)\left(1+\frac{\tau}{b^{p_2}}\right)}>0. \tag{46}$$

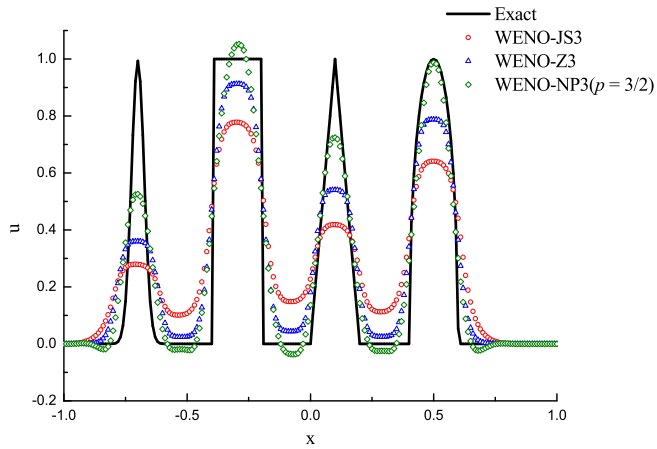
From Eq. (46), we can obtain the relation

$$\left[\frac{\omega_D}{\omega_C}\right]_{p_1}>\left[\frac{\omega_D}{\omega_C}\right]_{p_2}, (0<p_1<p_2<1). \tag{47}$$

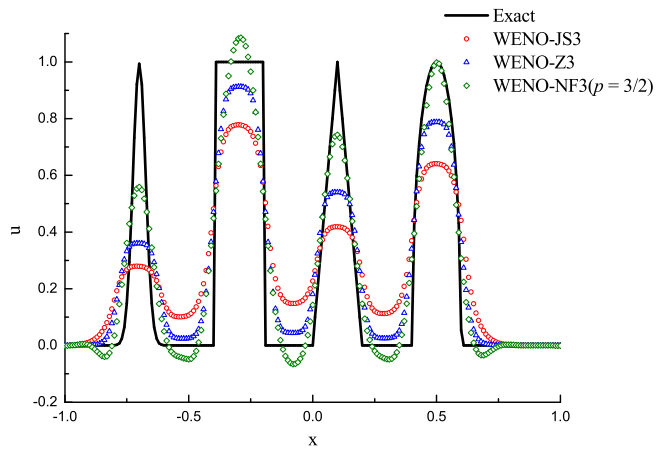
From Eq. (47), we can see that the WENO-PZ3 scheme will become less dissipative with smaller value of parameter p, which is also consistent with the above simulation results. However, as all know, the WENO schemes were designed to assign



(a) WENO-PZ3.



(b) WENO-NP3.



(c) WENO-NF3.

Fig. 1. Numerical solution of the advection equation with the discontinuous initial condition (40) for WENO-PZ3, WENO-NP3, and WENO-NF3 schemes.

very small weights to discontinuous substencils in order to capture the shock waves. Therefore, the less-smooth substencils weights should be kept small enough while raising them to some extent, for avoiding obvious spurious oscillations and instabilities appearing in the above two test with $p = 1/2$.

From above analysis, the parameter p in the WENO-PZ3 scheme was finally determined to be 3/4 considering balance between resolving power (dissipation) and the convergence order at the first-order critical point.

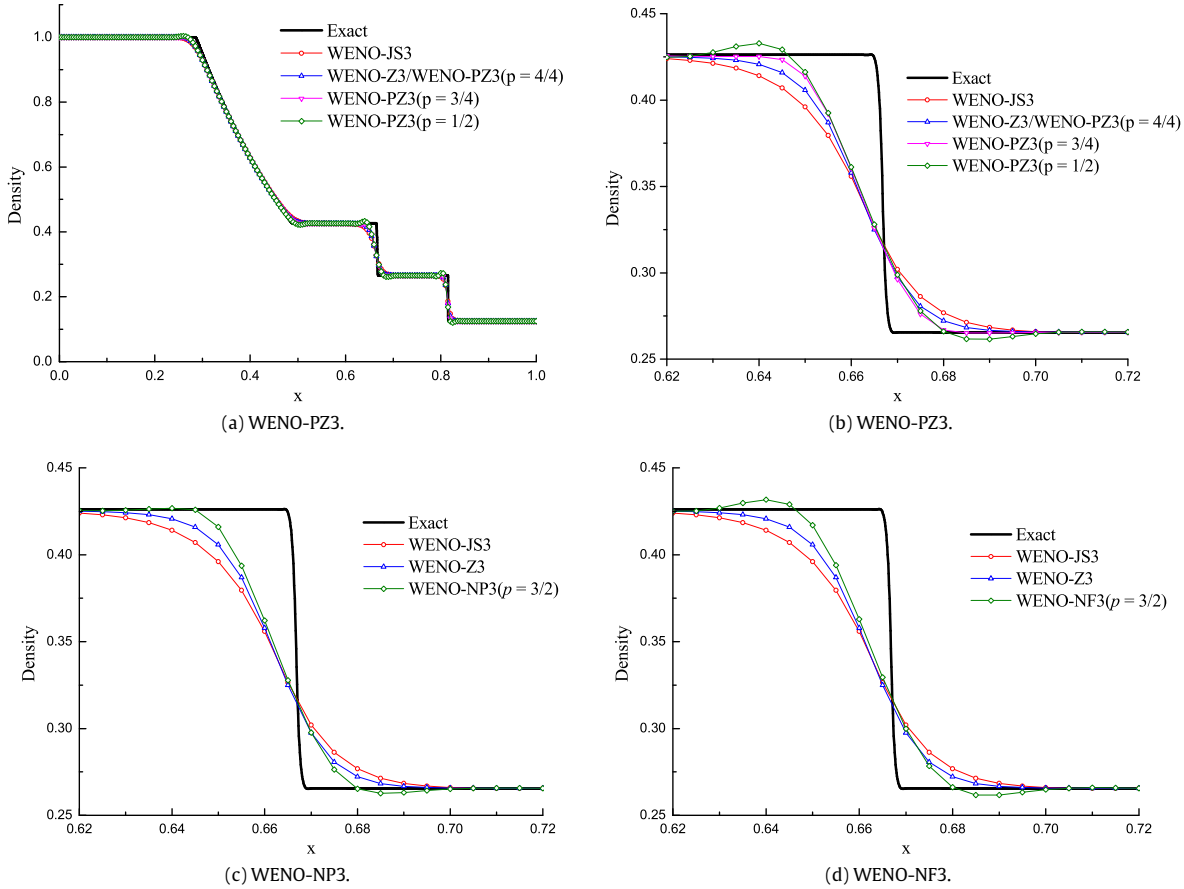


Fig. 2. Sod problem: The density and enlarged portion near the discontinuity for WENO-PZ3, WENO-NP3, and WENO-NF3 schemes.

The WENO-PZ3 scheme with the final parameter $p = 3/4$ achieves 2d-order accuracy at the critical points according to Eq. (38) while achieving 3d-order accuracy at the non-critical points in the smooth flow fields proved in the following Section 2.3.3. It should be noted here that the WENO-PZ3 scheme with the parameter $p = 3/4$ really performs well in the following numerical example in one-dimensional and two-dimensional space.

2.3.3. Accuracy of the improved scheme at non-critical points

Here, we will give a short proof that the WENO-PZ3 scheme with the parameter $p = 3/4$ can still achieves 3d-order accuracy at the non-critical points in the smooth flow fields with the way of Taylor expansion. Substituting Eqs. (18) and (31) into Eq. (30) gives

$$\begin{aligned}\alpha_0 &= d_0\left(1+\frac{\tau_Z}{\beta_0^p}\right)\\ &= d_0\left[1+2\left(f_i'\right)^{1-2p}f_i''h^{3-2p}+2p\left(f_i'\right)^{-2p}\left(f_i''\right)^2h^{4-2p}+O\left(h^{5-2p}\right)\right]\end{aligned} \tag{48}$$

$$\begin{aligned}\alpha_1 &= d_1\left(1+\frac{\tau_Z}{\beta_1^p}\right)\\ &= d_1\left[1+2\left(f_i'\right)^{1-2p}f_i''h^{3-2p}-2p\left(f_i'\right)^{-2p}\left(f_i''\right)^2h^{4-2p}+O\left(h^{5-2p}\right)\right].\end{aligned} \tag{49}$$

Then, with the weighting procedure in Eq. (5), the following relation can be obtained

$$\omega_0=\frac{\alpha_0}{\alpha_0+\alpha_1}=\frac{1}{3}+\frac{8}{9}p\left(f_i'\right)^{-2p}\left(f_i''\right)^2h^{4-2p}+O\left(h^{5-2p}\right) \tag{50}$$

$$\omega_1=\frac{\alpha_1}{\alpha_0+\alpha_1}=\frac{2}{3}-\frac{8}{9}p\left(f_i'\right)^{-2p}\left(f_i''\right)^2h^{4-2p}+O\left(h^{5-2p}\right). \tag{51}$$

Thus the numerical flux $\bar{f}_{i+1/2}$ in Eq. (2) is given by

$$\bar{f}_{i+1/2} = f_i + \frac{1}{2}f_i' h + \frac{1}{12}f_i'' h^2 + \frac{1}{12}f_i''' h^3 + \frac{1}{144}f_i^{(4)} h^4 - \frac{4}{9}p\left(f_i'\right)^{-2p}\left(f_i''\right)^3 h^{6-2p} + O\left(h^5\right). \tag{52}$$

Similarly, the numerical flux $\bar{f}_{i-1/2}$ in Eq. (2) is given by

$$\bar{f}_{i-1/2} = f_i - \frac{1}{2}f_i' h + \frac{1}{12}f_i'' h^2 + \frac{1}{12}f_i''' h^3 - \frac{11}{144}f_i^{(4)} h^4 - \frac{4}{9}p\left(f_i'\right)^{-2p}\left(f_i''\right)^3 h^{6-2p} + O\left(h^5\right). \tag{53}$$

Finally we get the following results with the parameter $p = 3/4$

$$\frac{\bar{f}_{i+1/2} - \bar{f}_{i-1/2}}{h} = f_i' + \frac{1}{12}f_i^{(4)} h^3 + O\left(h^4\right) = f_i' + O\left(h^3\right). \tag{54}$$

In fact, from Eqs. (52) and (53), it is clear that any value of the parameter p within the scope of [0, 1] can result in the same conclusion in Eq. (54).

2.4. Time marching methods

In this paper, Shu's 3rd order TVD time stepping method [2] is applied for the unsteady time integrations. The 3rd order TVD Runge–Kutta method is written as:

$$\begin{cases} u^{(1)} = u^n + \Delta t L\left(u^n\right) \\ u^{(2)} = \frac{3}{4}u^n + \frac{1}{4}u^{(1)} + \frac{1}{4}\Delta t L\left(u^{(1)}\right) \\ u^{n+1} = \frac{1}{3}u^n + \frac{1}{3}u^{(2)} + \frac{2}{3}\Delta t L\left(u^{(2)}\right). \end{cases} \tag{55}$$

3. Numerical results

In this section, we consider several one- and two-dimensional benchmark problems to illustrate the improvement produced by the present scheme.

3.1. Linear transport equation

The linear transport problems are controlled by

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0, -1 < x < 1 \qquad u(x, 0) = u_0(x), \textit{periodic}. \tag{56}$$

3.1.1. Critical point problem 1

A smooth initial condition,

$$u_0(x) = \sin(\pi x) \tag{57}$$

is taken in this case to verify the order of convergence at $t = 2$. In Table 1, the L_1, L_2, and L_∞ errors and the numerical orders of accuracy are given for WENO-JS3, WENO-Z3, and WENO-PZ3 schemes for this initial condition. It is noted that the WENO-PZ3 scheme achieves the third-order accuracy at the critical points as well as the WENO-JS3 scheme, which agrees well with the theoretical analysis in Section 2. It should be mentioned that the parameter ε has great influence on the accuracy WENO-JS3 scheme and It is demonstrated in [14,15] that WENO-JS3 scheme will locally degenerate to first order near critical point with $\varepsilon = 10^{-40}$.

3.1.2. Critical point problem 2

In this case, the initial condition in [3] is chosen as follows

$$u_0(x) = \sin\left(\pi x - \frac{\sin(\pi x)}{\pi}\right) \tag{58}$$

Table 1
A comparison study of L_1, L_2, and L_∞ (error and order) for linear advection equation with initial condition (57) at $t = 2$.

N	WENO-JS3		WENO-Z3		WENO-PZ3	
	L_1 (error)	L_1 (order)	L_1 (error)	L_1 (order)	L_1 (error)	L_1 (order)
25	1.0940E−1	–	5.3631E−2	–	2.3269E−2	–
50	4.2855E−2	1.3521	1.3589E−2	1.9806	2.7594E−3	3.0760
100	1.0946E−2	1.9691	3.0222E−3	2.1688	5.5014E−4	2.3265
200	1.0655E−3	3.3608	6.3660E−4	2.2471	6.9413E−5	2.9865
400	1.0565E−4	3.3342	1.3238E−4	2.2657	7.7655E−6	3.1601
	L_2 (error)	L_2 (order)	L_2 (error)	L_2 (order)	L_2 (error)	L_2 (order)
25	1.2102E−1	–	6.1653E−2	–	2.9668E−2	–
50	4.8961E−2	1.3055	1.8958E−2	1.7014	4.5515E−3	2.7045
100	1.6031E−2	1.6108	5.3617E−3	1.8220	1.1095E−3	2.0364
200	2.1026E−3	2.9306	1.4573E−3	1.8794	1.6747E−4	2.7279
400	2.4902E−4	3.0778	3.8955E−4	1.9034	1.7184E−5	3.2848
	L_∞ (error)	L_∞ (order)	L_∞ (error)	L_∞ (order)	L_∞ (error)	L_∞ (order)
25	2.0304E−1	–	1.2596E−1	–	7.1521E−2	–
50	8.7024E−2	1.2223	4.3521E−2	1.5332	1.3633E−2	2.3913
100	3.5593E−2	1.2898	1.6513E−2	1.3981	5.0061E−3	1.4453
200	7.2996E−3	2.2857	5.8361E−3	1.5005	9.0270E−4	2.4714
400	1.1783E−3	2.6311	2.0412E−3	1.5156	1.2590E−4	2.8420

Table 2
A comparison study of L_1, L_2, and L_∞ (error and order) for linear advection equation with initial condition (58) at $t = 2$.

N	WENO-JS3		WENO-Z3		WENO-PZ3	
	L_1 (error)	L_1 (order)	L_1 (error)	L_1 (order)	L_1 (error)	L_1 (order)
25	1.2429E−1	–	5.7276E−2	–	3.1239E−2	–
50	4.6050E−2	1.4324	1.5116E−2	1.9219	4.4240E−3	2.8199
100	1.3010E−2	1.8236	3.4645E−3	2.1254	5.7218E−4	2.9508
200	1.5132E−3	3.1039	8.3721E−4	2.0490	6.6413E−5	3.1069
400	1.4618E−4	3.3718	1.6823E−4	2.3152	8.2304E−6	3.0124
	L_2 (error)	L_2 (order)	L_2 (error)	L_2 (order)	L_2 (error)	L_2 (order)
25	1.4073E−1	–	6.8120E−2	–	3.9658E−2	–
50	5.7027E−2	1.3032	2.1294E−2	1.6776	7.0651E−3	2.4888
100	1.9545E−2	1.5448	6.1154E−3	1.7999	1.0032E−3	2.8161
200	2.9579E−3	2.7242	1.8892E−3	1.6947	1.1891E−4	3.0767
400	3.5424E−4	3.0618	4.9636E−4	1.9283	1.5992E−5	2.8944
	L_∞ (error)	L_∞ (order)	L_∞ (error)	L_∞ (order)	L_∞ (error)	L_∞ (order)
25	2.6012E−1	–	1.4836E−1	–	9.5066E−2	–
50	1.1014E−1	1.2398	5.6752E−2	1.3864	2.7095E−2	1.8109
100	4.6113E−2	1.2561	2.0758E−2	1.4510	5.0944E−3	2.4110
200	1.0219E−2	2.1739	7.4819E−3	1.4722	6.4780E−4	2.9753
400	1.6899E−3	2.5962	2.5623E−3	1.5460	1.2470E−4	2.3771

which contains two first-order critical points at which $f' = 0$ and $f''' \neq 0$. The L_1, L_2, and L_∞ errors along with the numerical order of accuracy are given in Table 2 for WENO-JS3, WENO-Z3, and WENO-PZ3 schemes. The proposed WENO-PZ3 scheme achieves the desired third order of accuracy.

3.1.3. Critical point problem 3

In this case [16], the initial condition is given by

$$u_0(x) = \sin^2(\pi x) \tag{59}$$

which contains a first-order critical point, that is, $f' = 0$ but $f'' \neq 0$. Table 3 shows the L_1, L_2, and L_∞ errors along with the numerical order of accuracy for WENO-JS3, WENO-Z3, and WENO-PZ3 schemes. The WENO-PZ3 scheme achieves good accuracy compared with other schemes.

3.1.4. Critical point problem 4

In this case [16], the initial condition is as follows

$$u_0(x) = \sin^3(\pi x) \tag{60}$$

which contains a first-order critical point, that is, $f' = 0$, $f'' = 0$, but $f''' \neq 0$. Table 4 shows the L_1, L_2, and L_∞ errors along with the numerical order of accuracy for WENO-JS3, WENO-Z3, and WENO-PZ3 schemes. In this case, the proposed WENO-PZ3 scheme performs better in comparison with WENO-JS3, WENO-Z3 schemes as well.

Table 3
A comparison study of L_1, L_2, and L_∞ (error and order) for linear advection equation with initial condition (59) at $t = 2$.

N	WENO-JS3		WENO-Z3		WENO-PZ3	
	L_1 (error)	L_1 (order)	L_1 (error)	L_1 (order)	L_1 (error)	L_1 (order)
25	2.3979E−1	–	1.5449E−1	–	1.0460E−1	–
50	9.7096E−2	1.3043	4.0085E−2	1.9464	1.4229E−2	2.8780
100	3.7344E−2	1.3785	1.1356E−2	1.8196	2.2190E−3	2.6809
200	8.0248E−3	2.2183	2.5484E−3	2.1558	3.5290E−4	2.6526
400	6.6808E−4	3.5864	5.2704E−4	2.2736	3.4763E−5	3.3436
	L_2 (error)	L_2 (order)	L_2 (error)	L_2 (order)	L_2 (error)	L_2 (order)
25	2.5795E−1	–	1.7237E−1	–	1.1800E−1	–
50	1.0755E−1	1.2621	4.8112E−2	1.8410	1.9563E−2	2.5926
100	4.2041E−2	1.3551	1.4957E−2	1.6856	3.7353E−3	2.3889
200	1.0986E−2	1.9361	4.1485E−3	1.8502	6.3334E−4	2.5602
400	1.2208E−3	3.1698	1.1154E−3	1.8950	7.0924E−5	3.1586
	L_∞ (error)	L_∞ (order)	L_∞ (error)	L_∞ (order)	L_∞ (error)	L_∞ (order)
25	3.6523E−1	–	2.7085E−1	–	1.9601E−1	–
50	1.7035E−1	1.1003	9.0203E−2	1.5862	4.2772E−2	2.1962
100	6.8022E−2	1.3244	3.2841E−2	1.4577	1.1473E−2	1..8984
200	2.2665E−2	1.5855	1.1653E−2	1.4948	2.2716E−3	2.3365
400	3.9230E−3	2.5304	4.0814E−3	1.5136	3.5380E−4	2.6827

Table 4
A comparison study of L_1, L_2, and L_∞ (error and order) for linear advection equation with initial condition (60) at $t = 2$.

N	WENO-JS3		WENO-Z3		WENO-PZ3	
	L_1 (error)	L_1 (order)	L_1 (error)	L_1 (order)	L_1 (error)	L_1 (order)
25	1.8499E−1	–	1.5221E−1	–	1.2291E−1	–
50	9.1142E−2	1.0213	4.1257E−2	1.8834	2.2940E−2	2.4217
100	2.9843E−2	1.6107	9.1524E−3	2.1724	3.1623E−3	2.8588
200	7.3790E−3	2.0159	2.0387E−3	2.1665	4.7473E−4	2.7358
400	7.0804E−4	3.3815	4.1500E−4	2.2965	5.4892E−5	3.1124
	L_2 (error)	L_2 (order)	L_2 (error)	L_2 (order)	L_2 (error)	L_2 (order)
25	2.0722E−1	–	1.6906E−1	–	1.3646E−1	–
50	1.0844E−1	0.9343	5.1808E−2	1.7063	2.7722E−2	2.2994
100	4.0613E−2	1.4169	1.5325E−2	1.7573	4.7100E−3	2.5572
200	1.2684E−2	1.6789	4.2739E−3	1.8423	8.5761E−4	2.4573
400	1.5831E−3	3.0022	1.1588E−3	1.8829	9.9995E−5	3.1004
	L_∞ (error)	L_∞ (order)	L_∞ (error)	L_∞ (order)	L_∞ (error)	L_∞ (order)
25	3.8696E−1	–	3.1669E−1	–	2.5060E−1	–
50	2.2025E−1	0.8130	1.2435E−1	1.3487	6.7881E−2	1.8843
100	9.7458E−2	1.1763	4.8963E−2	1.3446	1.9052E−2	1.8331
200	3.6946E−2	1.3994	1.7450E−2	1.4885	4.9018E−3	1.9586
400	7.1187E−3	2.3757	6.1192E−3	1.5118	6.6310E−4	2.8860

3.1.5. *Linear discontinuity problem*

A solution with an initial discontinuity [4] is calculated

$$u_0(x) = \begin{cases} -\sin(\pi x) - \frac{1}{2}x^3, & -1 < x \le 0 \\ -\sin(\pi x) - \frac{1}{2}x^3 + 1, & 0 < x \le 1. \end{cases} \tag{61}$$

The numerical solutions for the WENO-JS3, WENO-Z3, and WENO-PZ3 schemes with grid number $N = 200$ at $t = 6$ are plotted in Fig. 3. It is obvious that, near the discontinuity, the WENO-PZ3 scheme obtains sharper solution than the other schemes. And, near the critical points, the WENO-PZ3 scheme gives more accurate results.

3.1.6. *Complex wave*

A more complex initial solution is tested in [11]

$$u_0(x) = \begin{cases} -x\sin\left(3\pi x^2/2\right), & -1 \le x < -1/3 \\ |\sin(2\pi x)|, & -1/3 \le x < 1/3 \\ 2x - 1 - \sin(3\pi x)/6, & otherwise. \end{cases} \tag{62}$$

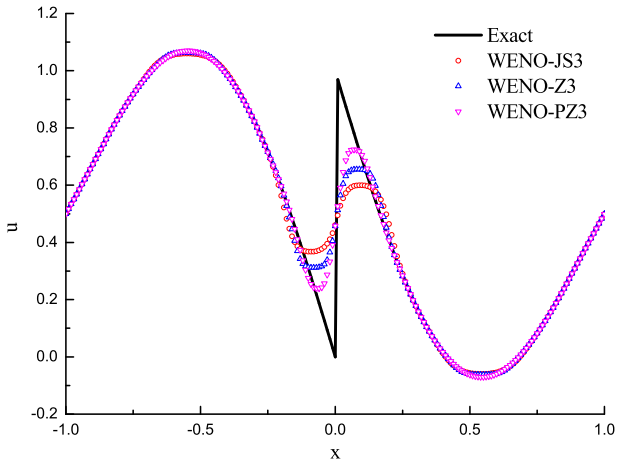


Fig. 3. Numerical results with the initial solution (61), $t = 6$.

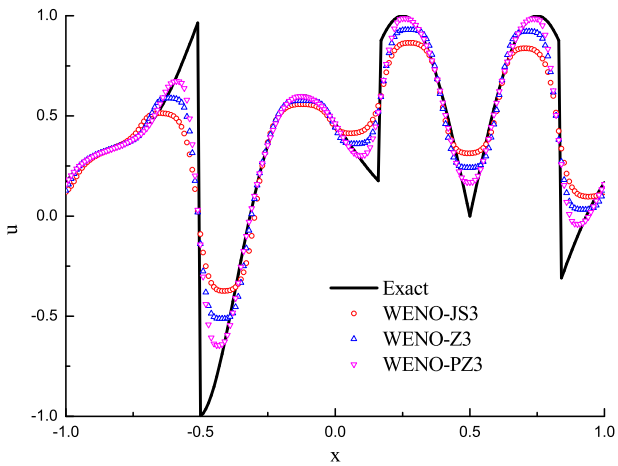


Fig. 4. Numerical results with the initial solution (62), $t = 6$.

Fig. 4 plots the numerical solutions with 200 grid points at $t = 6$. It can be seen that the WENO-PZ3 scheme achieves noticeably higher amplitudes near discontinuities and more accuracy results at critical points than WENO-JS3 and WENO-Z3 schemes.

3.2. One-dimensional shock tube problems

In this section, we present numerical examples with the one dimensional system of the Euler equations for gas dynamics in strong conservation form:

$$\frac{\partial \boldsymbol{U}}{\partial t} + \frac{\partial \boldsymbol{F}}{\partial x} = 0 \tag{63}$$

where

$$\boldsymbol{U} = \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix}, \boldsymbol{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u\,(E + p) \end{bmatrix}, p = (\gamma - 1)\left(E - \frac{1}{2}\rho u^2\right). \tag{64}$$

And ρ, u, p, and E are the density, velocity, pressure and total energy respectively. γ is the ratio of specific heats with the value of 1.4.

3.2.1. Shock–entropy wave test of Shu–Osher

The initial condition for the 1D Shu–Osher problem [18] is given by

$$(\rho, u, p) = \begin{cases} (3.857143,\, 2.629369,\, 10.33333)\,, & -5 \le x < -4 \\ (1 + 0.2\sin 5x,\, 0,\, 1)\,, & -4 \le x \le 5. \end{cases} \tag{65}$$

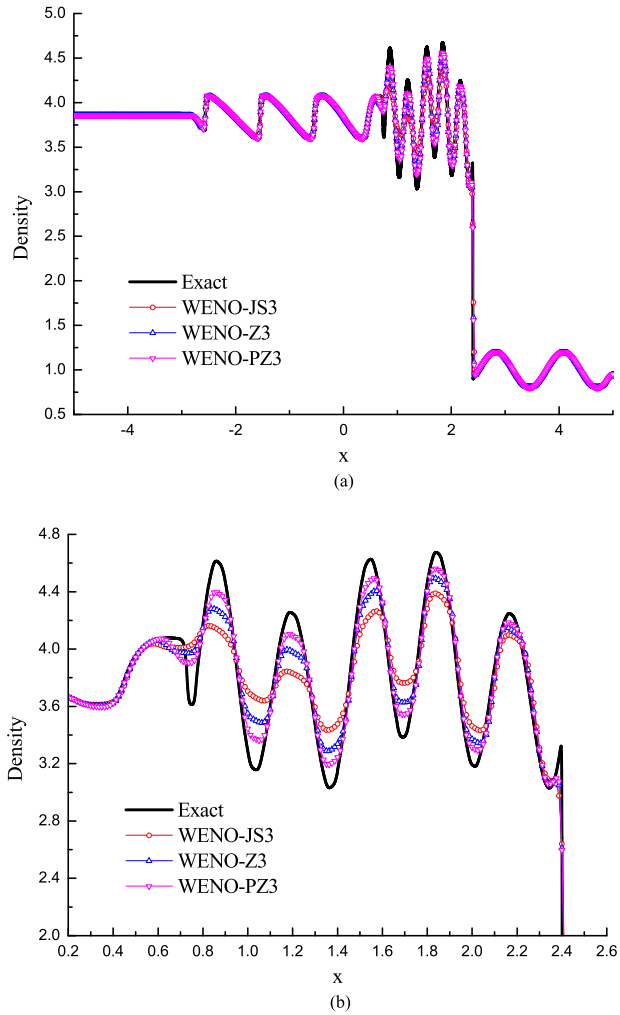


Fig. 5. Shu–Osher problem: (a) density distribution and (b) the enlarged portion of the high frequency waves region.

Fig. 5 displays a comparison for all schemes with $N = 800$ points at $t = 1.8$. The reference solutions are computed by the WENO-JS3 scheme with grid points of $N = 10000$. It can be seen that the WENO-PZ3 scheme captures the finer scale structures at the high-frequency waves behind the shock than WENO-JS3 and WENO-Z3 schemes.

3.2.2. Shock–entropy wave test of Titarev–Toro

A variation of the Shu–Osher test with a different initial condition [19] is given as follows

$$(\rho, u, p) = \begin{cases} (1.515695, 0.523346, 1.805000)\,, & -5 \le x < -4.5 \\ (1 + 0.1\sin(20\pi x)\,, 0, 1)\,, & -4.5 \le x \le 5. \end{cases} \tag{66}$$

The flow contains physical oscillations which have to be resolved by the numerical method. We compute the solution until the output time $t = 5$. Fig. 6 depicts a comparison of results for all schemes on a mesh of 4000 cells. The reference solution is obtained by applying the WENO-PZ3 scheme on a fine mesh of 6000 cells. From Fig. 6, it is obvious that the WENO-PZ3 scheme resolves most of the waves with a good approximation to their amplitudes better over other tested schemes.

3.3. Two-dimensional Euler problems

In this section, we apply the WENO schemes to several complex flow simulations in a two-dimensional space. The governing equations are the two-dimensional Euler equations in Cartesian coordinates given by

$$\frac{\partial \boldsymbol{U}}{\partial t} + \frac{\partial \boldsymbol{F}}{\partial x} + \frac{\partial \boldsymbol{G}}{\partial x} = 0 \tag{67}$$

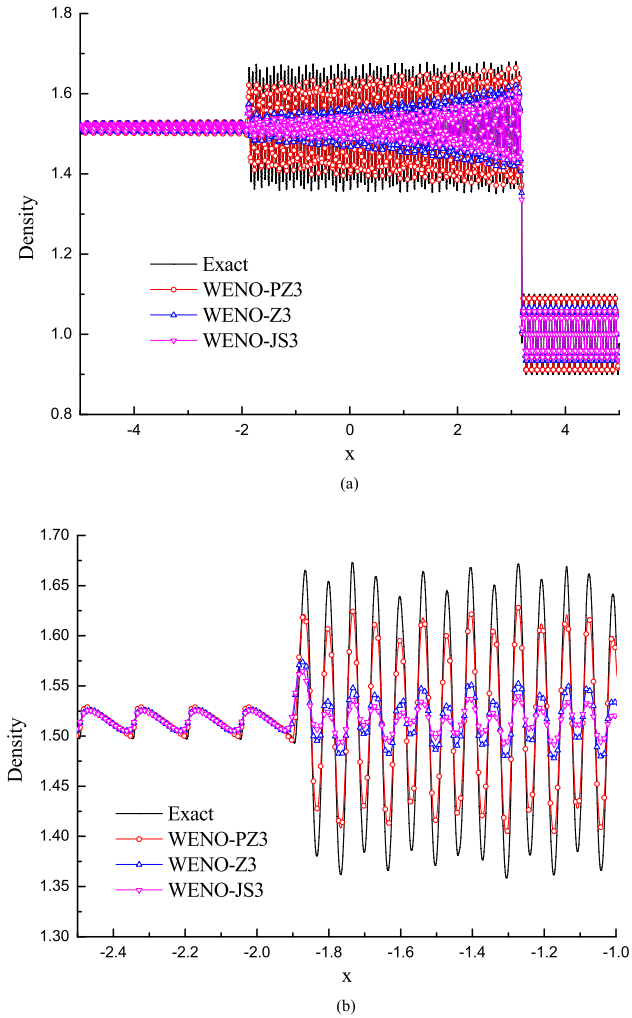


Fig. 6. Titarev–Toro problem: (a) density distribution and (b) the enlarged portion of the high frequency waves region.

where

$$\boldsymbol{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E \end{pmatrix} \boldsymbol{F} = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u\,(E + p) \end{pmatrix} \boldsymbol{G} = \begin{pmatrix} \rho v \\ \rho v u \\ \rho v^2 + p \\ v\,(E + p) \end{pmatrix}. \tag{68}$$

And the equation of state is

$$p = (\gamma - 1)\left(E - \frac{1}{2}\rho u^2 - \frac{1}{2}\rho v^2\right) \tag{69}$$

where ρ, u, v, p, and E are the density, x-velocity, y-velocity, pressure and total energy respectively. γ is the ratio of specific heats.

3.3.1. Two-dimensional Riemann problem

The problem [20] is solved on the square $[0, 1] \times [0, 1]$, which is divided into four quadrants by lines $x = 0.8$ and $y = 0.8$. The Riemann problems are defined by initial constant states in each quadrant as follows:

$$\begin{aligned} &(\rho, u, v, p) = (1.5, 0, 0, 1.5)\,, && 0.8 \le x \le 1, 0.8 \le y \le 1 \\ &(\rho, u, v, p) = (0.5323, 1.206, 0, 0.3)\,, && 0 \le x < 0.8, 0.8 \le y \le 1 \\ &(\rho, u, v, p) = (0.138, 1.206, 1.206, 0.029)\,, && 0 \le x < 0.8, 0 \le y < 0.8 \\ &(\rho, u, v, p) = (0.5323, 0, 1.206, 0.3)\,, && 0.8 < x \le 1, 0 \le y < 0.8. \end{aligned} \tag{70}$$

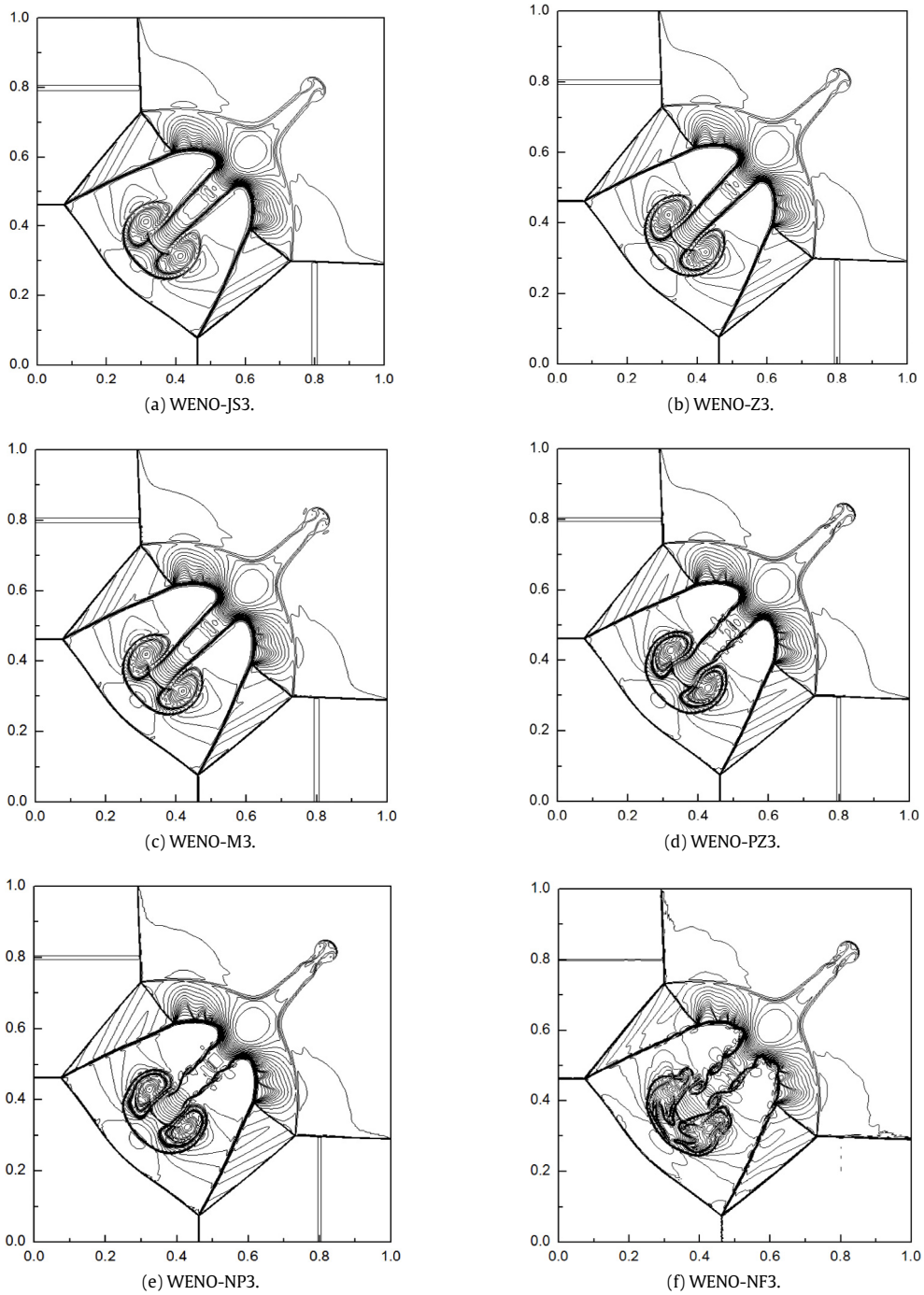
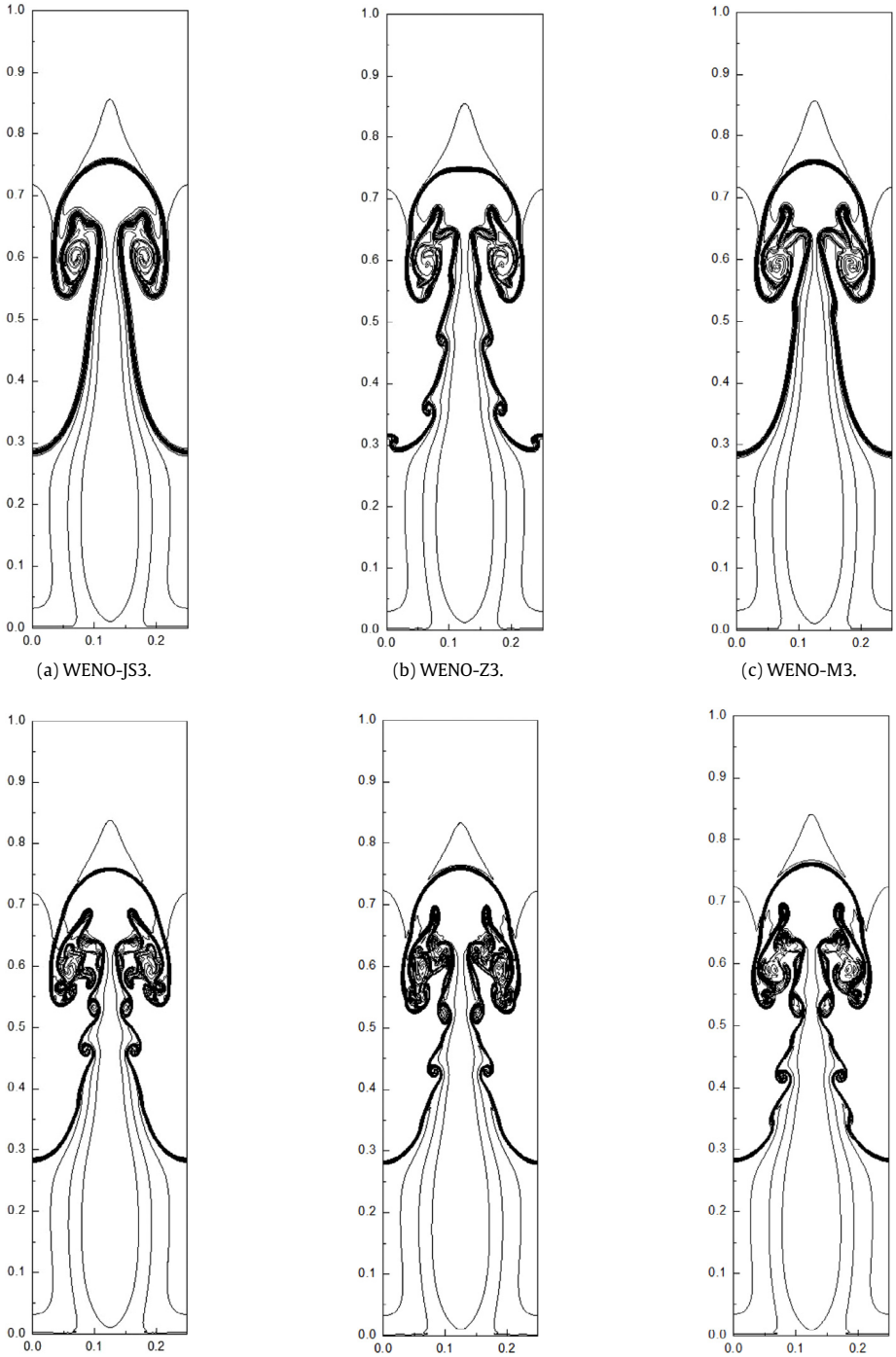


Fig. 7. Density contours of two-dimensional Riemann problem computed by the WENO-JS3, WENO-Z3, WENO-M3, WENO-PZ3, WENO-NP3, and WENO-NF3 schemes at $t = 0.8$. 40 contours were fit between the range of 0.14 to 1.7.

We evolve this initial data until time $t = 0.8$. The ratio of specific heats is $\gamma = 1.4$. Fig. 7 depicts the results of WENO-JS3, WENO-Z3, WENO-M3 (Applying the mapping function in [3] to the WENO-JS3 scheme), WENO-PZ3, WENO-NP3, and WENO-NF3 on the computational region of 960×960 grid points. Careful examination of these results reveals that WENO-PZ3 presents a higher resolution of the structure of discrete vortex along the slip line than WENO-JS3 and WENO-Z3 schemes. It is also indicated from the simulation results that WENO-NP3 and the WENO-NF3 schemes perform better than the WENO-PZ3 scheme. In this case, there are little differences between the WENO-Z3 and WENO-M3 schemes.



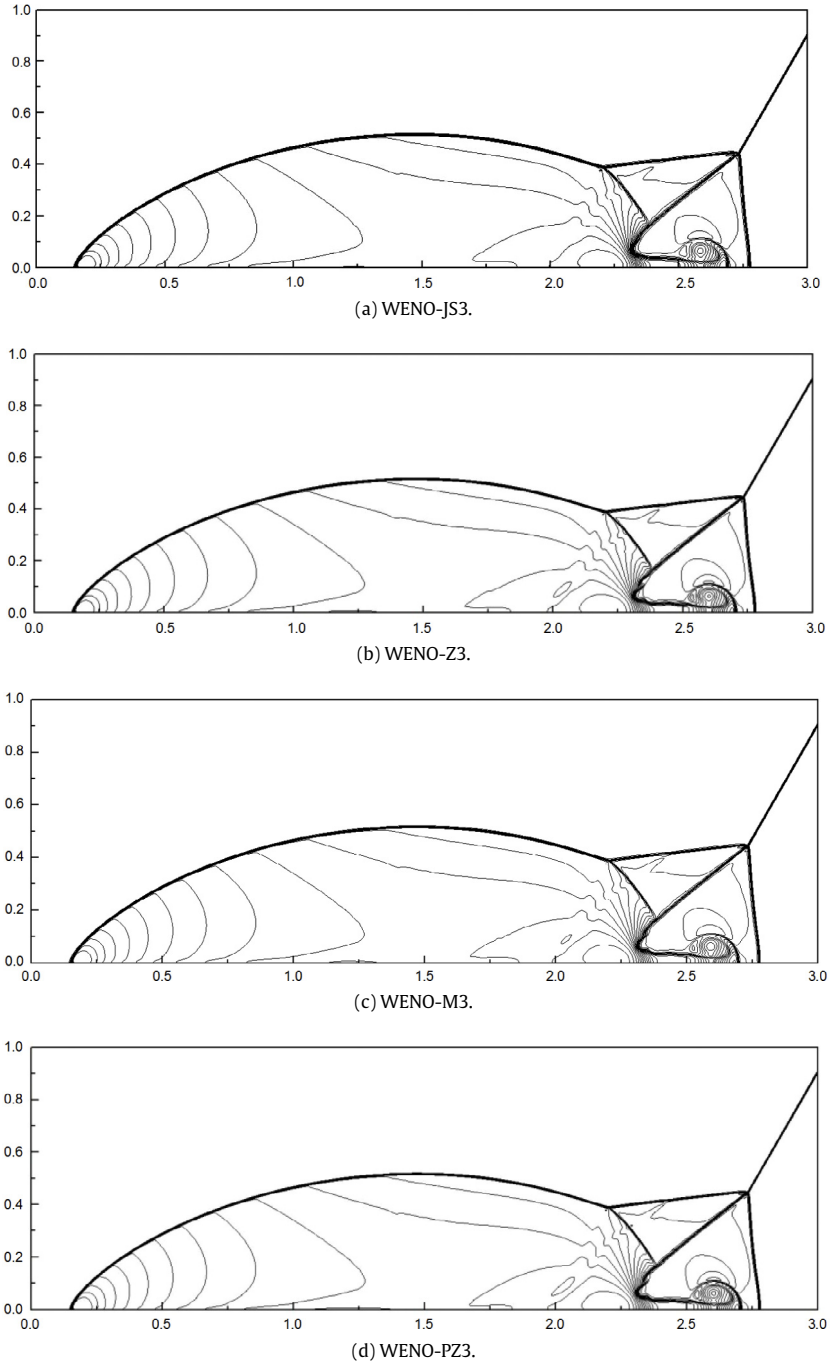
(a) WENO-JS3. (b) WENO-Z3. (c) WENO-M3.

(d) WENO-PZ3. (e) WENO-NP3. (f) WENO-NF3.

Fig. 8. Density contours of the Rayleigh–Taylor instability computed by the WENO-JS3, WENO-Z3, WENO-M3, WENO-PZ3, WENO-NP3, and WENO-NF3 schemes, and 15 contours were fit between the range of 0.952269 to 2.14589.

3.3.2. Rayleigh–Taylor instability

This problem has been simulated extensively by many authors [14,16,17,21,22] to assess the numerical dissipations of the numerical schemes. It happens on an interface instability between different densities when an acceleration is directed from heavy fluid to light one. The computational domain is $[0, 0.25] \times [0, 1]$, and the initial conditions are



(a) WENO-JS3.

(b) WENO-Z3.

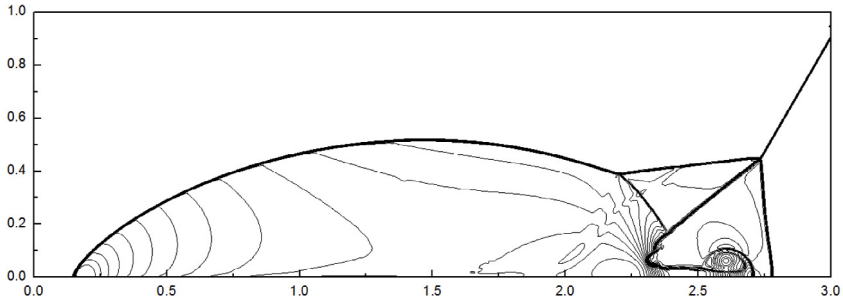
(c) WENO-M3.

(d) WENO-PZ3.

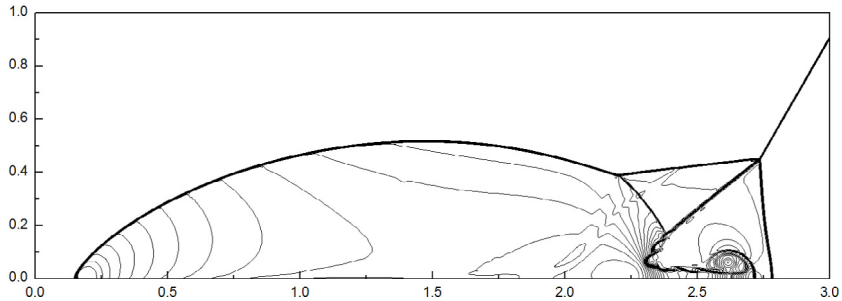
Fig. 9. Double Mach reflection computed by the WENO-JS3, WENO-Z3, WENO-M3, WENO-PZ3, WENO-NP3, and WENO-NF3 schemes from the top to bottom at $t = 0.2$, and 40 equally spaced contour lines from 1.8 to 20.

$$(\rho, u, v, p) = \begin{cases} \left(2, 0, -0.025\sqrt{\gamma p/\rho}\cos(8\pi x), 2y+1\right), & 0 \le y < 0.5 \\ \left(1, 0, -0.025\sqrt{\gamma p/\rho}\cos(8\pi x), y+1.5\right), & 0.5 \le y < 1. \end{cases} \tag{71}$$

The gravity effect is considered by adding the source terms ρ to the right hand side of the y-momentum equation and ρv to the energy equation. The ratio of specific heats γ is set to be 5/3, and reflective boundary conditions are imposed for the left and right boundaries; The top boundary are set as $(\rho, u, v, p) = (1, 0, 0, 2.5)$ and $(\rho, u, v, p) = (2, 0, 0, 1)$ are set to be the bottom boundary condition. The final time is $t = 1.95$.



(e) WENO-NP3.



(f) WENO-NF3.

Fig. 9. (*continued*)

Fig. 8 displays the density contours of the solutions computed on the 240×960 uniform grid. It is obvious from Fig. 8 that the contact discontinuity resolution (manifested by the small scale structures) is significantly improved by the WENO-PZ3 scheme. The WENO-PZ3, WENO-NP3, WENO-NF3 schemes give the similar results. It is clear in this case that the WENO-Z3 scheme performs better than the WENO-M3 scheme.

3.3.3. Double Mach reflection of a strong shock

This problem describes the phenomenon that a Mach 10 shock reflected from the wall with an incidence angle of 60°, which is initially proposed and studied in detail by Woodward and Colella [23]. The initial conditions are

$$(\rho, u, v, p) = \begin{cases} (8.0, 7.145, -4.125, 116.5)\,, & x < \dfrac{1}{6} + \dfrac{y}{\sqrt{3}} \\ (1.4, 0, 0, 1)\,, & x \geq \dfrac{1}{6} + \dfrac{y}{\sqrt{3}}. \end{cases} \tag{72}$$

The ratio of specific heats is $\gamma = 1.4$. We run this computation till $t = 0.2$. The results in $[0, 3] \times [0, 1]$ are displayed. Fig. 9 shows the results of WENO-JS3, WENO-Z3, WENO-M3, WENO-PZ3, WENO-NP3, and WENO-NF3 on a computation region with 1920×480 grid points. Partial enlarged details of the region around the double Mach stems are shown in Fig. 10. It can be seen that the present WENO-PZ3 scheme resolves the instabilities little better around the Mach stem than the WENO-Z3 scheme. The WENO-NP3 scheme shows almost the same results as the WENO-PZ3, while the WENO-NF3 gives the best results among the above tested schemes. What is more, the WENO-Z3 scheme performs little better than the WENO-M3 scheme.

4. Conclusions

In this paper, we have introduced an improved third-order WENO-Z scheme by slightly modifying the local smoothness indicators of the conventional WENO-Z scheme. The final parameter is determined to be 3/4 considering the balance between resolving power (dissipation) and the convergence order at the first-order critical point. It is observed that the WENO-PZ3 scheme performs better compared with WENO-JS3, WENO-Z3 schemes by several one- and two-dimensional benchmark problems owing to the combined effects of the low dissipation and convergence order improvement.

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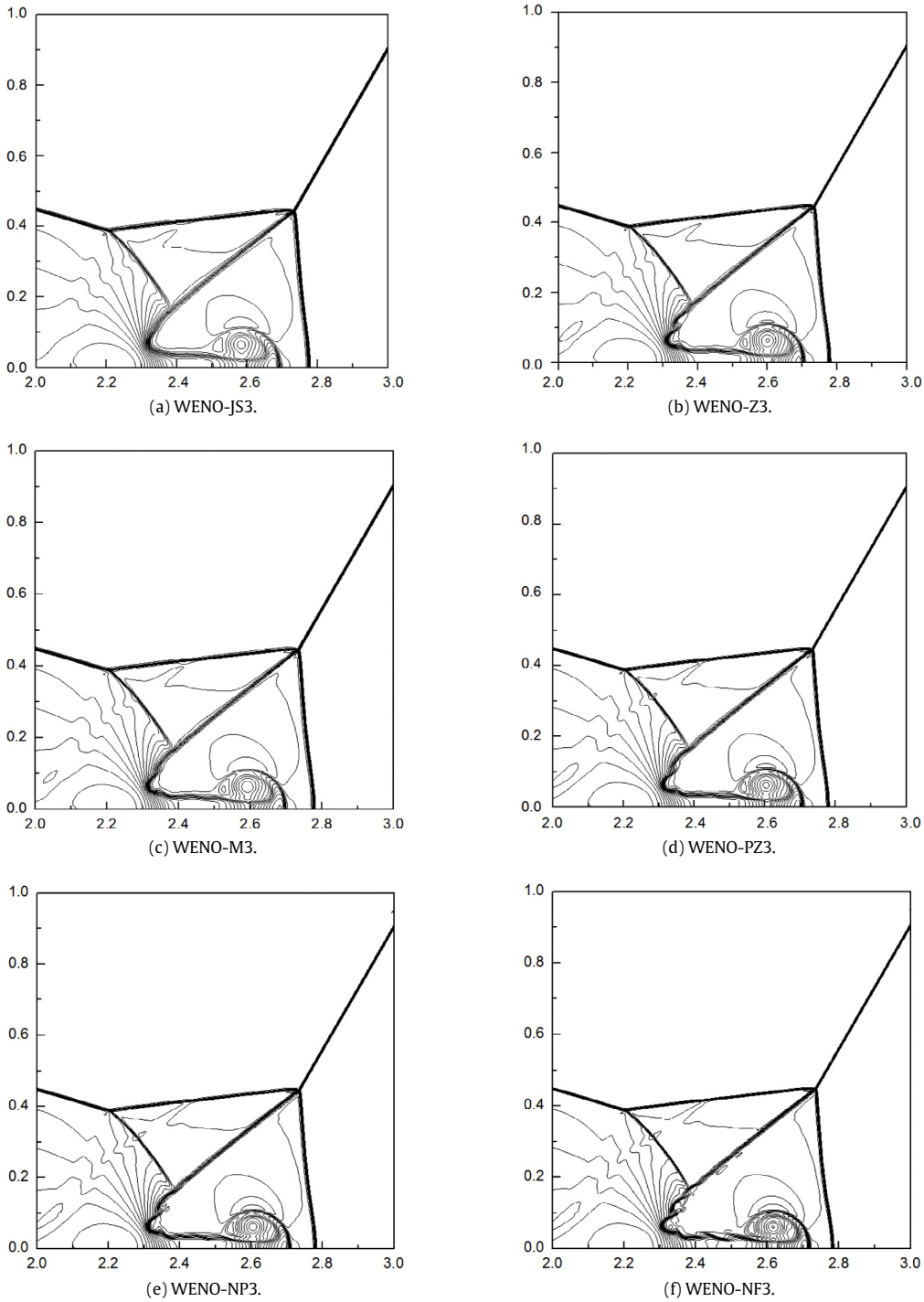

(a) WENO-JS3. (b) WENO-Z3.

(c) WENO-M3. (d) WENO-PZ3.

(e) WENO-NP3. (f) WENO-NF3.

Fig. 10. Density contours of double Mach stems computed by the WENO-JS3, WENO-Z3, WENO-M3, WENO-PZ3, WENO-NP3, and WENO-NF3 schemes at $t = 0.2$.

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Appendix

We think the method in the present work can also be applied to high-order scheme. Here, we apply this method to the fifth order WENO scheme. The following are the detailed derivations:

(1) *WENO-JS5 scheme*

The fifth-order WENO scheme [2] (WENO-JS5) use a convex combination of the candidate numerical fluxes to reconstruct the final flux

$$\bar{f}_{i+1/2} = \omega_0 f_{0,i+1/2} + \omega_1 f_{1,i+1/2} + \omega_2 f_{2,i+1/2} \tag{73}$$

where $f_{0,i+1/2}, f_{1,i+1/2}, f_{2,i+1/2}$ are the third-order fluxes defined on the three stencils $\{x_{i-2}, x_{i-1}, x_i\}$, $\{x_{i-1}, x_i, x_{i+1}\}$, $\{x_i, x_{i+1}, x_{i+2}\}$ as follows:

$$\begin{aligned} f_{0,i+1/2} &= \frac{1}{3}f_{i-2} - \frac{7}{6}f_{i-1} + \frac{11}{6}f_i, \\ f_{1,i+1/2} &= -\frac{1}{6}f_{i-1} + \frac{5}{6}f_i + \frac{1}{3}f_{i+1}, \\ f_{2,i+1/2} &= \frac{1}{3}f_i + \frac{5}{6}f_{i+1} - \frac{1}{6}f_{i+2}. \end{aligned} \tag{74}$$

The classical weight functions ω_0, ω_1 and ω_2 proposed by Jiang and Shu in [2] are given by

$$\omega_k = \frac{\alpha_k}{\sum_{s=0}^{1}\alpha_s}, \alpha_k = \frac{d_k}{(\varepsilon + \beta_k)^2}, k = 0, 1, 2 \tag{75}$$

where the parameter ε in Eq. (75) is set to be 10^{-6} in the present work, as recommended in [2]. The optimal weights d_k, with the values

$$d_0 = \frac{1}{10}, d_1 = \frac{6}{10}, d_2 = \frac{3}{10}. \tag{76}$$

The smoothness indicators are as follows

$$\begin{cases} \beta_0 = \frac{13}{12}(f_{i-2} - 2f_{i-1} + f_i)^2 + \frac{1}{4}(f_{i-2} - 4f_{i-1} + 3f_i)^2 \\ \beta_1 = \frac{13}{12}(f_{i-1} - 2f_i + f_{i+1})^2 + \frac{1}{4}(f_{i-1} - f_{i+1})^2 \\ \beta_2 = \frac{13}{12}(f_i - 2f_{i+1} + f_{i+2})^2 + \frac{1}{4}(3f_i - 4f_{i+1} + f_{i+2})^2. \end{cases} \tag{77}$$

(2) *The improved WENO-Z5 scheme*

For the fifth-order WENO-Z scheme (WENO-Z5 scheme), the function α_k proposed in [4] is defined as

$$\alpha_k = d_k\left(1 + \frac{\tau_Z}{\beta_k + \varepsilon}\right), k = 0, 1, 2 \tag{78}$$

$$\tau_Z = |\beta_0 - \beta_2|, \varepsilon = 10^{-40}. \tag{79}$$

Thus, the improved WENO-Z5 scheme can be written as the following form

$$\alpha_k = d_k\left(1 + \frac{\tau_Z}{(\beta_k + \varepsilon)^p}\right), k = 0, 1, 2 \tag{80}$$

$$\tau_Z = |\beta_0 - \beta_2|, \varepsilon = 10^{-40}. \tag{81}$$

At the first-order critical point ($f_i' = 0, f_i'' \neq 0, f_i''' \neq 0$), Eq. (77) can be written as the following form disregarding the small parameter ε

$$\begin{aligned} \beta_0 = &\frac{13}{12}f_i''^2h^4 - \frac{13}{6}f_i''f_i'''h^5 + \frac{91}{72}f_i''f_i^{(4)}h^6 + \frac{43}{36}f_i'''^2h^6 \\ &- \frac{13}{24}f_i''f_i^{(5)}h^7 - \frac{103}{72}f_i'''f_i^{(4)}h^7 + O\left(h^8\right) \end{aligned} \tag{82}$$

$$\beta_1 = \frac{13}{12}f_i''^2h^4 + \frac{13}{72}f_i''f_i^{(4)}h^6 + \frac{1}{36}f_i'''^2h^6 + O\left(h^8\right) \tag{83}$$

$$\begin{aligned} \beta_2 = &\frac{13}{12}f_i''^2h^4 + \frac{13}{6}f_i''f_i'''h^5 + \frac{91}{72}f_i''f_i^{(4)}h^6 + \frac{43}{36}f_i'''^2h^6 \\ &+ \frac{13}{24}f_i''f_i^{(5)}h^7 + \frac{103}{72}f_i'''f_i^{(4)}h^7 + O\left(h^8\right). \end{aligned} \tag{84}$$

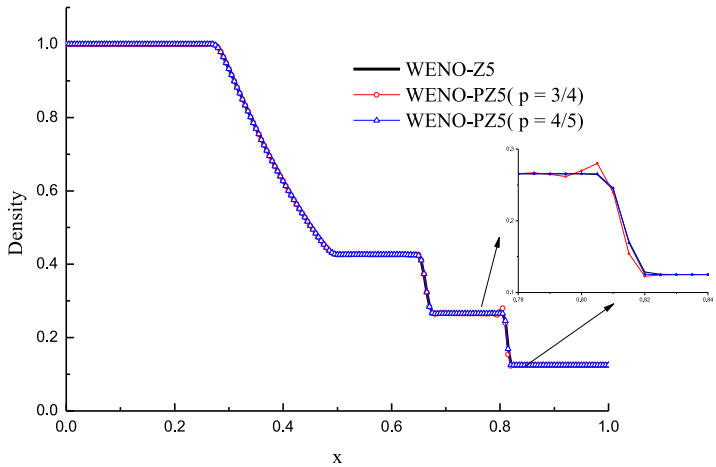

Fig. 11. Comparisons between WENO-Z5 and WENO-PZ5 schemes for Sod problem.

Substituting Eqs. (82)–(84), and Eq. (81) into Eq. (80) gives

$$\alpha_0 = d_0\left(1 + \frac{\frac{13}{3}f_i''f_i'''}{\left(\frac{13}{12}f_i''^2\right)^p}h^{5-4p}\left(1 + 2p\frac{f_i'''}{f_i''}h + O\left(h^2\right)\right)\right) \tag{85}$$

$$\alpha_1 = d_1\left(1 + \frac{\frac{13}{3}f_i''f_i'''}{\left(\frac{13}{12}f_i''^2\right)^p}h^{5-4p}\left(1 + O\left(h^2\right)\right)\right) \tag{86}$$

$$\alpha_2 = d_2\left(1 + \frac{\frac{13}{3}f_i''f_i'''}{\left(\frac{13}{12}f_i''^2\right)^p}h^{5-4p}\left(1 - 2p\frac{f_i'''}{f_i''}h + O\left(h^2\right)\right)\right). \tag{87}$$

Then, with the weighting procedure in Eq. (75), we can obtain the following relation

$$\omega_0 = \frac{\alpha_0}{\alpha_0+\alpha_1+\alpha_2} = d_0\left(1 + \frac{\frac{13}{3}f_i''f_i'''}{\left(\frac{13}{12}f_i''^2\right)^p}\frac{12}{5}p\frac{f_i'''}{f_i''}h^{6-4p} + O\left(h^{7-4p}\right)\right) \tag{88}$$

$$\omega_1 = \frac{\alpha_1}{\alpha_0+\alpha_1+\alpha_2} = d_1\left(1 + \frac{\frac{13}{3}f_i''f_i'''}{\left(\frac{13}{12}f_i''^2\right)^p}\frac{2}{5}p\frac{f_i'''}{f_i''}h^{6-4p} + O\left(h^{7-4p}\right)\right) \tag{89}$$

$$\omega_2 = \frac{\alpha_2}{\alpha_0+\alpha_1+\alpha_2} = d_2\left(1 - \frac{\frac{13}{3}f_i''f_i'''}{\left(\frac{13}{12}f_i''^2\right)^p}\frac{8}{5}p\frac{f_i'''}{f_i''}h^{6-4p} + O\left(h^{7-4p}\right)\right). \tag{90}$$

Thus the numerical flux $\bar{f}_{i+1/2}$ in Eq. (73) is given by

$$\bar{f}_{i+1/2} = f_i + \frac{1}{2}f_i'h + \frac{1}{12}f_i''h^2 - \frac{1}{720}f_i^{(4)}h^4 - \frac{1}{60}f_i^{(5)}h^5 + O\left(h^6\right). \tag{91}$$

Similarly, the numerical flux $\bar{f}_{i-1/2}$ is given by

$$\begin{aligned}\bar{f}_{i-1/2} &= f_i - \frac{1}{2}f_i'h + \frac{1}{12}f_i''h^2 - \frac{1}{720}f_i^{(4)}h^4 - \frac{1}{60}f_i^{(5)}h^5 \\ &\quad + \frac{3}{125}p\frac{\frac{13}{3}f_i''f_i'''}{\left(\frac{25}{12}f_i''^2\right)^p}\frac{f_i''f_i'''}{f_i'}h^{9-4p} + O\left(h^6\right).\end{aligned} \tag{92}$$

Finally we get the following results

$$\frac{\bar{f}_{i+1/2} - \bar{f}_{i-1/2}}{h} = f_i' - \frac{3}{125}p\frac{\frac{13}{3}f_i''f_i'''}{\left(\frac{25}{12}f_i''^2\right)^p}\frac{f_i''f_i'''}{f_i'}h^{8-4p} + O\left(h^5\right). \tag{93}$$

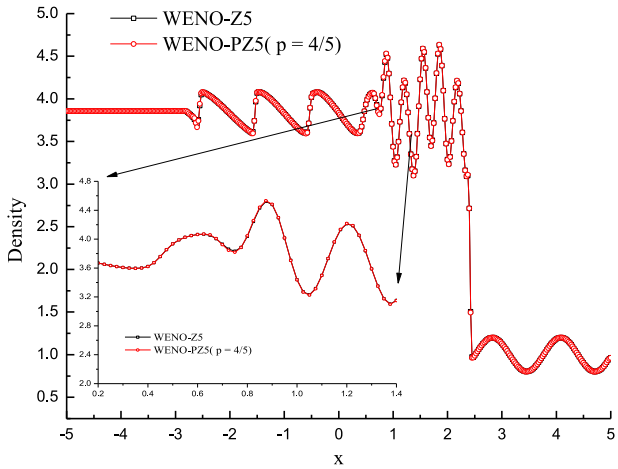

Fig. 12. Comparisons between WENO-Z5 and WENO-PZ5 schemes for Shu–Osher problem.

From Eq. (93), we can find that the parameter p should satisfy the following condition to achieve the optimal order of convergence at the first-order critical points

$$8 - 4p \geq 5 \Rightarrow p \leq \frac{3}{4}. \tag{94}$$

For convenience, this proposed fifth-order WENO-Z scheme is denoted by WENO-PZ5 scheme. Then, we choose the Sod shock tube case seen in Eq. (42) to determine the final parameter p (shown in Fig. 11). From Fig. 11, we can find that the WENO-PZ5 scheme with the parameter $p = 3/4$ gives obvious spurious oscillations. When we slightly increase the value of parameter p to 4/5, it can give good results. So, the final parameter p is finally determined to be 4/5.

It is also indicated from Fig. 11 that, the WENO-PZ5 scheme with the parameter $p = 4/5$ just shows little improvements. What is more, we also conduct the simulation of Shu–Osher problem as shown in Fig. 12.

From Fig. 12, we find that the WENO-PZ5 scheme only shows little improvement than the WENO-Z5 scheme. Thus, we think that the present method is especially suitable for the third-order WENO schemes. Of course, the present scheme WENO-PZ5 really performs better than the conventional WENO-JS5 scheme and it also worth spreading.

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